

Class 14: RNA-Seq Analysis Mini-Project

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Section 1: Differential Expression Analysis

```
library(DESeq2)
```

```
Loading required package: S4Vectors
```

```
Loading required package: stats4
```

```
Loading required package: BiocGenerics
```

```
Loading required package: generics
```

```
Attaching package: 'generics'
```

```
The following objects are masked from 'package:base':
```

```
as.difftime, as.factor, as.ordered, intersect, is.element, setdiff,  
setequal, union
```

Attaching package: 'BiocGenerics'

The following objects are masked from 'package:stats':

IQR, mad, sd, var, xtabs

The following objects are masked from 'package:base':

anyDuplicated, aperm, append, as.data.frame, basename, cbind,
colnames, dirname, do.call, duplicated, eval, evalq, Filter, Find,
get, grep, grepl, is.unsorted, lapply, Map, mapply, match, mget,
order, paste, pmax, pmax.int, pmin, pmin.int, Position, rank,
rbind, Reduce, rownames, sapply, saveRDS, table, tapply, unique,
unsplit, which.max, which.min

Attaching package: 'S4Vectors'

The following object is masked from 'package:utils':

findMatches

The following objects are masked from 'package:base':

expand.grid, I, unname

Loading required package: IRanges

Loading required package: GenomicRanges

Loading required package: Seqinfo

Loading required package: SummarizedExperiment

Loading required package: MatrixGenerics

Loading required package: matrixStats

Attaching package: 'MatrixGenerics'

The following objects are masked from 'package:matrixStats':

colAlls, colAnyNAs, colAnys, colAvgPerRowSet, colCollapse,
colCounts, colCummaxs, colCummins, colCumprods, colCumsums,
colDiffs, colIQRDiffs, colIQRs, colLogSumExps, colMadDiffs,
colMads, colMaxs, colMeans2, colMedians, colMins, colOrderStats,
colProds, colQuantiles, colRanges, colRanks, colSdDiffs, colSds,
colSums2, colTabulates, colVarDiffs, colVars, colWeightedMads,
colWeightedMeans, colWeightedMedians, colWeightedSds,
colWeightedVars, rowAlls, rowAnyNAs, rowAnys, rowAvgPerColSet,
rowCollapse, rowCounts, rowCummaxs, rowCummins, rowCumprods,
rowCumsums, rowDiffs, rowIQRDiffs, rowIQRs, rowLogSumExps,
rowMadDiffs, rowMads, rowMaxs, rowMeans2, rowMedians, rowMins,
rowOrderStats, rowProds, rowQuantiles, rowRanges, rowRanks,
rowSdDiffs, rowSds, rowSums2, rowTabulates, rowVarDiffs, rowVars,
rowWeightedMads, rowWeightedMeans, rowWeightedMedians,
rowWeightedSds, rowWeightedVars

Loading required package: Biobase

Welcome to Bioconductor

Vignettes contain introductory material; view with
'browseVignettes()'. To cite Bioconductor, see
'citation("Biobase")', and for packages 'citation("pkgname")'.

Attaching package: 'Biobase'

The following object is masked from 'package:MatrixGenerics':

rowMedians

The following objects are masked from 'package:matrixStats':

anyMissing, rowMedians

```
metaFile <- "GSE37704_metadata.csv"
countFile <- "GSE37704_featurecounts.csv"

colData = read.csv(metaFile, row.names=1)
head(colData)
```

```
      condition
SRR493366 control_sirna
SRR493367 control_sirna
SRR493368 control_sirna
SRR493369      hoxa1_kd
SRR493370      hoxa1_kd
SRR493371      hoxa1_kd
```

```
countData = read.csv(countFile, row.names=1)
head(countData)
```

```
      length SRR493366 SRR493367 SRR493368 SRR493369 SRR493370
ENSG00000186092    918         0         0         0         0
ENSG00000279928    718         0         0         0         0
ENSG00000279457   1982        23        28        29        28
ENSG00000278566    939         0         0         0         0
ENSG00000273547    939         0         0         0         0
ENSG00000187634   3214       124       123       205       207
                                SRR493371
ENSG00000186092         0
ENSG00000279928         0
ENSG00000279457        46
ENSG00000278566         0
ENSG00000273547         0
ENSG00000187634       258
```

```
colnames(countData)
```

```
[1] "length"      "SRR493366" "SRR493367" "SRR493368" "SRR493369" "SRR493370"
[7] "SRR493371"
```

```
rownames(colData)
```

```
[1] "SRR493366" "SRR493367" "SRR493368" "SRR493369" "SRR493370" "SRR493371"
```

Q1. Complete the code below to remove the troublesome first column from countData

```
countData <- as.matrix(countData[, -1])
head(countData)
```

	SRR493366	SRR493367	SRR493368	SRR493369	SRR493370	SRR493371
ENSG00000186092	0	0	0	0	0	0
ENSG00000279928	0	0	0	0	0	0
ENSG00000279457	23	28	29	29	28	46
ENSG00000278566	0	0	0	0	0	0
ENSG00000273547	0	0	0	0	0	0
ENSG00000187634	124	123	205	207	212	258

Q2. Complete the code below to filter countData to exclude genes (i.e. rows) where we have 0 read count across all samples (i.e. columns).

```
countData <- countData[rowSums(countData) > 0, ]
head(countData)
```

	SRR493366	SRR493367	SRR493368	SRR493369	SRR493370	SRR493371
ENSG00000279457	23	28	29	29	28	46
ENSG00000187634	124	123	205	207	212	258
ENSG00000188976	1637	1831	2383	1226	1326	1504
ENSG00000187961	120	153	180	236	255	357
ENSG00000187583	24	48	65	44	48	64
ENSG00000187642	4	9	16	14	16	16

```
dds = DESeqDataSetFromMatrix(countData=countData,
                              colData=colData,
                              design=~condition)
```

Warning in DESeqDataSet(se, design = design, ignoreRank): some variables in design formula are characters, converting to factors

```
dds = DESeq(dds)
```

estimating size factors

estimating dispersions

gene-wise dispersion estimates

mean-dispersion relationship

final dispersion estimates

fitting model and testing

```
dds
```

```
class: DESeqDataSet
dim: 15975 6
metadata(1): version
assays(4): counts mu H cooks
rownames(15975): ENSG00000279457 ENSG00000187634 ... ENSG00000276345
               ENSG00000271254
rowData names(22): baseMean baseVar ... deviance maxCooks
colnames(6): SRR493366 SRR493367 ... SRR493370 SRR493371
colData names(2): condition sizeFactor
```

```
res = results(dds)
```

Q3. Call the `summary()` function on your results to get a sense of how many genes are up or down-regulated at the default 0.1 p-value cutoff.

```
summary(res)
```

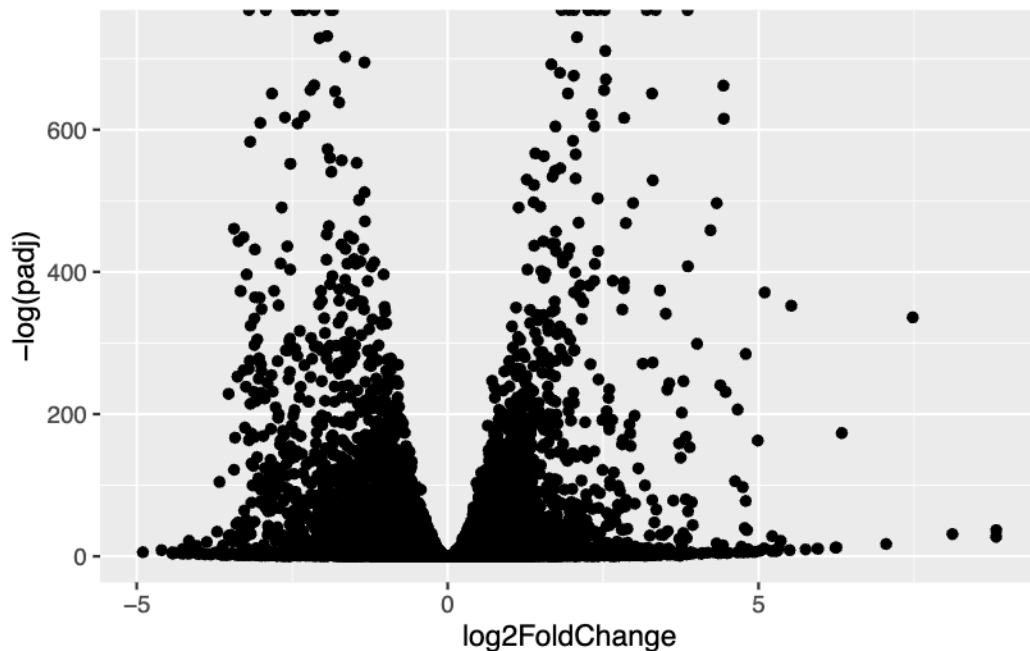
```
out of 15975 with nonzero total read count
adjusted p-value < 0.1
LFC > 0 (up)      : 4349, 27%
LFC < 0 (down)    : 4396, 28%
outliers [1]      : 0, 0%
low counts [2]    : 1237, 7.7%
(mean count < 0)
[1] see 'cooksCutoff' argument of ?results
[2] see 'independentFiltering' argument of ?results
```

The `summary(res)` function shows the number of significantly upregulated and downregulated genes at the adjusted p-value cutoff of 0.1. And in these results, 4349 genes were upregulated and 4396 genes were downregulated.

```
library(ggplot2)

ggplot(res) +
  aes(log2FoldChange,
      -log(padj)) +
  geom_point()
```

Warning: Removed 1237 rows containing missing values or values outside the scale range (``geom_point()``).



Q4. Improve this plot by completing the below code, which adds color, axis labels and cutoff lines:

```
mycols <- rep("gray", nrow(res))

# genes meet both the thresholds
inds <- (res$padj < 0.05) &
  (abs(res$log2FoldChange) > 2)
```

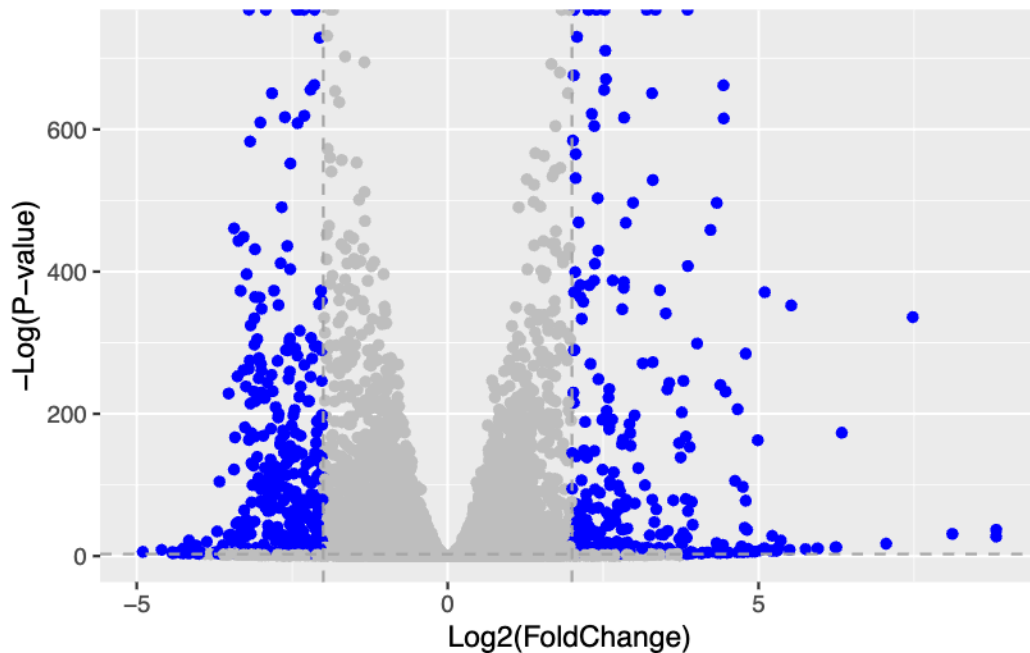
```

mycols[inds] <- "blue"

ggplot(res) +
  aes(log2FoldChange,
      -log(padj)) +
  geom_point(col=mycols) +
  xlab("Log2(FoldChange)") +
  ylab("-Log(P-value)") +
  geom_vline(xintercept=c(-2,2),
             col="darkgray",
             lty=2) +
  geom_hline(yintercept=-log(0.05),
             col="darkgray",
             lty=2)

```

Warning: Removed 1237 rows containing missing values or values outside the scale range (`geom\_point()`).



Q5. Use the `mapIDs()` function multiple times to add SYMBOL, ENTREZID and GENENAME annotation to our results by completing the code below.


```
library("AnnotationDbi")
library("org.Hs.eg.db")
```

```
columns(org.Hs.eg.db)
```

```
[1] "ACCNUM"      "ALIAS"        "ENSEMBL"      "ENSEMBLPROT"  "ENSEMBLTRANS"
[6] "ENTREZID"    "ENZYME"       "EVIDENCE"     "EVIDENCEALL"  "GENENAME"
[11] "GENETYPE"    "GO"           "GOALL"        "IPI"           "MAP"
[16] "OMIM"        "ONTOLOGY"     "ONTOLOGYALL"  "PATH"          "PFAM"
[21] "PMID"        "PROSITE"      "REFSEQ"       "SYMBOL"        "UCSCKG"
[26] "UNIPROT"
```

```
res$symbol = mapIds(org.Hs.eg.db,
                     keys=row.names(res),
                     keytype="ENSEMBL",
                     column="SYMBOL",
                     multiVals="first")
```

'select()' returned 1:many mapping between keys and columns

```
res$entrez = mapIds(org.Hs.eg.db,
                     keys=row.names(res),
                     keytype="ENSEMBL",
                     column="ENTREZID",
                     multiVals="first")
```

'select()' returned 1:many mapping between keys and columns

```
res$name = mapIds(org.Hs.eg.db,
                   keys=row.names(res),
                   keytype="ENSEMBL",
                   column="GENENAME",
                   multiVals="first")
```

'select()' returned 1:many mapping between keys and columns

Q6. Finally for this section let's reorder these results by adjusted p-value and save them to a CSV file in your current project directory.

```
res = res[order(res$padj),]  
write.csv(res, file="deseq_results.csv")
```

Section 2: Pathway Analysis

```
library(pathview)
```

```
#####  
Pathview is an open source software package distributed under GNU General  
Public License version 3 (GPLv3). Details of GPLv3 is available at  
http://www.gnu.org/licenses/gpl-3.0.html. Particullary, users are required to  
formally cite the original Pathview paper (not just mention it) in publications  
or products. For details, do citation("pathview") within R.
```

The pathview downloads and uses KEGG data. Non-academic uses may require a KEGG license agreement (details at <http://www.kegg.jp/kegg/legal.html>).

```
#####
```

```
library(gage)
```

```
library(gageData)  
  
data(kegg.sets.hs)  
data(sigmet.idx.hs)  
  
# focus on signaling and metabolic pathways only  
kegg.sets.hs = kegg.sets.hs[sigmet.idx.hs]  
  
head(kegg.sets.hs, 3)
```

```
$`hsa00232 Caffeine metabolism`  
[1] "10"    "1544" "1548" "1549" "1553" "7498" "9"
```

```
$`hsa00983 Drug metabolism - other enzymes`
```

```
[1] "10"      "1066"    "10720"   "10941"   "151531"  "1548"    "1549"    "1551"
[9] "1553"    "1576"    "1577"    "1806"    "1807"    "1890"    "221223"  "2990"
[17] "3251"    "3614"    "3615"    "3704"    "51733"   "54490"   "54575"   "54576"
[25] "54577"   "54578"   "54579"   "54600"   "54657"   "54658"   "54659"   "54963"
[33] "574537"  "64816"   "7083"    "7084"    "7172"    "7363"    "7364"    "7365"
[41] "7366"    "7367"    "7371"    "7372"    "7378"    "7498"    "79799"   "83549"
[49] "8824"    "8833"    "9"       "978"
```

```
$`hsa00230 Purine metabolism`
```

```
[1] "100"      "10201"   "10606"   "10621"   "10622"   "10623"   "107"     "10714"
[9] "108"      "10846"   "109"     "111"     "11128"   "11164"   "112"     "113"
[17] "114"      "115"     "122481"  "122622"  "124583"  "132"     "158"     "159"
[25] "1633"     "171568"  "1716"    "196883"  "203"     "204"     "205"     "221823"
[33] "2272"     "22978"   "23649"   "246721"  "25885"   "2618"    "26289"   "270"
[41] "271"      "27115"   "272"     "2766"    "2977"    "2982"    "2983"    "2984"
[49] "2986"     "2987"    "29922"   "3000"    "30833"   "30834"   "318"     "3251"
[57] "353"      "3614"    "3615"    "3704"    "377841"  "471"     "4830"    "4831"
[65] "4832"     "4833"    "4860"    "4881"    "4882"    "4907"    "50484"   "50940"
[73] "51082"    "51251"   "51292"   "5136"    "5137"    "5138"    "5139"    "5140"
[81] "5141"     "5142"    "5143"    "5144"    "5145"    "5146"    "5147"    "5148"
[89] "5149"     "5150"    "5151"    "5152"    "5153"    "5158"    "5167"    "5169"
[97] "51728"    "5198"    "5236"    "5313"    "5315"    "53343"   "54107"   "5422"
[105] "5424"     "5425"    "5426"    "5427"    "5430"    "5431"    "5432"    "5433"
[113] "5434"     "5435"    "5436"    "5437"    "5438"    "5439"    "5440"    "5441"
[121] "5471"     "548644"  "55276"   "5557"    "5558"    "55703"   "55811"   "55821"
[129] "5631"     "5634"    "56655"   "56953"   "56985"   "57804"   "58497"   "6240"
[137] "6241"     "64425"   "646625"  "654364"  "661"     "7498"    "8382"    "84172"
[145] "84265"    "84284"   "84618"   "8622"    "8654"    "87178"   "8833"    "9060"
[153] "9061"     "93034"   "953"     "9533"    "954"     "955"     "956"     "957"
[161] "9583"     "9615"
```

```
foldchanges = res$log2FoldChange
names(foldchanges) = res$entrez
head(foldchanges)
```

```
1266      54855      1465      2034      2150      6659
-2.422719  3.201955 -2.313738 -1.888019  3.344508  2.392288
```

```
# obtain the results
keggres = gage(foldchanges, gsets=kegg.sets.hs)
```

```
attributes(keggres)
```

```
$names
```

```
[1] "greater" "less"    "stats"
```

```
# look at the first few down (less) pathways
```

```
head(keggres$less)
```

	p.geomean	stat.mean	p.val
hsa04110 Cell cycle	8.995727e-06	-4.378644	8.995727e-06
hsa03030 DNA replication	9.424076e-05	-3.951803	9.424076e-05
hsa03013 RNA transport	1.375901e-03	-3.028500	1.375901e-03
hsa03440 Homologous recombination	3.066756e-03	-2.852899	3.066756e-03
hsa04114 Oocyte meiosis	3.784520e-03	-2.698128	3.784520e-03
hsa00010 Glycolysis / Gluconeogenesis	8.961413e-03	-2.405398	8.961413e-03

	q.val	set.size	exp1
hsa04110 Cell cycle	0.001448312	121	8.995727e-06
hsa03030 DNA replication	0.007586381	36	9.424076e-05
hsa03013 RNA transport	0.073840037	144	1.375901e-03
hsa03440 Homologous recombination	0.121861535	28	3.066756e-03
hsa04114 Oocyte meiosis	0.121861535	102	3.784520e-03
hsa00010 Glycolysis / Gluconeogenesis	0.212222694	53	8.961413e-03

```
pathview(gene.data=foldchanges, pathway.id="hsa04110")
```

'select()' returned 1:1 mapping between keys and columns

Info: Working in directory /Users/melissabui/BIMM143/class14

Info: Writing image file hsa04110.pathview.png

```
# a different PDF based output of the same data
```

```
pathview(gene.data=foldchanges, pathway.id="hsa04110", kegg.native=FALSE)
```

'select()' returned 1:1 mapping between keys and columns

Warning: reconcile groups sharing member nodes!

```
      [,1] [,2]
[1,] "9"  "300"
[2,] "9"  "306"
```

Info: Working in directory /Users/melissabui/BIMM143/class14

Info: Writing image file hsa04110.pathview.pdf

```
## focus on top 5 upregulated pathways here for demo purposes
keggrespathways <- rownames(keggres$greater)[1:5]

# extract the 8 character long IDs part of each string
keggresids = substr(keggrespathways, start=1, stop=8)
keggresids
```

```
[1] "hsa04640" "hsa04630" "hsa00140" "hsa04142" "hsa04330"
```

```
pathview(gene.data=foldchanges, pathway.id=keggresids, species="hsa")
```

'select()' returned 1:1 mapping between keys and columns

Info: Working in directory /Users/melissabui/BIMM143/class14

Info: Writing image file hsa04640.pathview.png

'select()' returned 1:1 mapping between keys and columns

Info: Working in directory /Users/melissabui/BIMM143/class14

Info: Writing image file hsa04630.pathview.png

'select()' returned 1:1 mapping between keys and columns

Info: Working in directory /Users/melissabui/BIMM143/class14

Info: Writing image file hsa00140.pathview.png

'select()' returned 1:1 mapping between keys and columns

Info: Working in directory /Users/melissabui/BIMM143/class14

Info: Writing image file hsa04142.pathview.png

'select()' returned 1:1 mapping between keys and columns

Info: Working in directory /Users/melissabui/BIMM143/class14

Info: Writing image file hsa04330.pathview.png

Q7. Can you do the same procedure as above to plot the pathview figures for the top 5 down-regulated pathways?

```
keggrespathways_down <- rownames(keggres$less)[1:5]
keggresids_down = substr(keggrespathways_down, start=1, stop=8)
keggresids_down
```

```
[1] "hsa04110" "hsa03030" "hsa03013" "hsa03440" "hsa04114"
```

```
pathview(gene.data=foldchanges, pathway.id=keggresids_down, species="hsa")
```

'select()' returned 1:1 mapping between keys and columns

Info: Working in directory /Users/melissabui/BIMM143/class14

Info: Writing image file hsa04110.pathview.png

'select()' returned 1:1 mapping between keys and columns

Info: Working in directory /Users/melissabui/BIMM143/class14

Info: Writing image file hsa03030.pathview.png

'select()' returned 1:1 mapping between keys and columns

Info: Working in directory /Users/melissabui/BIMM143/class14

Info: Writing image file hsa03013.pathview.png

'select()' returned 1:1 mapping between keys and columns

Info: Working in directory /Users/melissabui/BIMM143/class14

Info: Writing image file hsa03440.pathview.png

'select()' returned 1:1 mapping between keys and columns

Info: Working in directory /Users/melissabui/BIMM143/class14

Info: Writing image file hsa04114.pathview.png

The `keggrespathways_down <- rownames(keggres$less)[1:5]` code extracts the top five downregulated KEGG pathways from `keggres$less`, as this pulls out their KEGG pathway IDs. Additionally we use `pathview()` function to generate pathway diagrams showing how genes in each pathway are upregulated or downregulated as a result.

Section 3: Gene Ontology (GO)

```
data(go.sets.hs)
data(go.subs.hs)

# focusing on Biological Process subset of GO
gobpsets = go.sets.hs[go.subs.hs$BP]

gobpres = gage(foldchanges, gsets=gobpsets)

lapply(gobpres, head)
```


\$greater

		p.geomean	stat.mean	p.val
G0:0007156	homophilic cell adhesion	8.519724e-05	3.824205	8.519724e-05
G0:0002009	morphogenesis of an epithelium	1.396681e-04	3.653886	1.396681e-04
G0:0048729	tissue morphogenesis	1.432451e-04	3.643242	1.432451e-04
G0:0007610	behavior	1.925222e-04	3.565432	1.925222e-04
G0:0060562	epithelial tube morphogenesis	5.932837e-04	3.261376	5.932837e-04
G0:0035295	tube development	5.953254e-04	3.253665	5.953254e-04
		q.val	set.size	exp1
G0:0007156	homophilic cell adhesion	0.1951953	113	8.519724e-05
G0:0002009	morphogenesis of an epithelium	0.1951953	339	1.396681e-04
G0:0048729	tissue morphogenesis	0.1951953	424	1.432451e-04
G0:0007610	behavior	0.1967577	426	1.925222e-04
G0:0060562	epithelial tube morphogenesis	0.3565320	257	5.932837e-04
G0:0035295	tube development	0.3565320	391	5.953254e-04

\$less

		p.geomean	stat.mean	p.val
G0:0048285	organelle fission	1.536227e-15	-8.063910	1.536227e-15
G0:0000280	nuclear division	4.286961e-15	-7.939217	4.286961e-15
G0:0007067	mitosis	4.286961e-15	-7.939217	4.286961e-15
G0:0000087	M phase of mitotic cell cycle	1.169934e-14	-7.797496	1.169934e-14
G0:0007059	chromosome segregation	2.028624e-11	-6.878340	2.028624e-11
G0:0000236	mitotic prometaphase	1.729553e-10	-6.695966	1.729553e-10
		q.val	set.size	exp1
G0:0048285	organelle fission	5.841698e-12	376	1.536227e-15
G0:0000280	nuclear division	5.841698e-12	352	4.286961e-15
G0:0007067	mitosis	5.841698e-12	352	4.286961e-15
G0:0000087	M phase of mitotic cell cycle	1.195672e-11	362	1.169934e-14
G0:0007059	chromosome segregation	1.658603e-08	142	2.028624e-11
G0:0000236	mitotic prometaphase	1.178402e-07	84	1.729553e-10

\$stats

		stat.mean	exp1
G0:0007156	homophilic cell adhesion	3.824205	3.824205
G0:0002009	morphogenesis of an epithelium	3.653886	3.653886
G0:0048729	tissue morphogenesis	3.643242	3.643242
G0:0007610	behavior	3.565432	3.565432
G0:0060562	epithelial tube morphogenesis	3.261376	3.261376
G0:0035295	tube development	3.253665	3.253665

The GO analysis identified a couple significantly biological processes related to immune response, signaling as well as cell cycle regulation. These results are also consistent with the KEGG

pathway analysis. This the idea that dex treatment strongly affects transcriptional regulation and cellular signaling pathways.

Section 4: Reactome Analysis

```
sig_genes <- res[res$padj <= 0.05 & !is.na(res$padj), "symbol"]  
print(paste("Total number of significant genes:", length(sig_genes)))
```

```
[1] "Total number of significant genes: 8147"
```

```
write.table(sig_genes, file="significant_genes.txt", row.names=FALSE, col.names=FALSE, quote=
```

Q8. What pathway has the most significant “Entities p-value”? Do the most significant pathways listed match your previous KEGG results? What factors could cause differences between the two methods?

The pathway that has the most significant “Entities p-value” is Reactome pathways, which were related to cell cycle regulation and DNA replication. These results were consistent with the earlier KEGG pathway analysis, where pathways such as Cell Cycle and DNA Replication were also strongly downregulated. The similarities suggest that HOXA1 knockdown strongly affects genes involved in cell cycle progression. As a result, the differences between Reactome and KEGG results could occur since the databases uses different pathway definitions, annotations, and gene mappings.